

CS & IT ENGINEERING



Computer Network

Flow Control

Lecture No. - 01



By - Abhishek Sir



Recap of Previous Lecture



Topic

Hamming Distance

Topic

Bit Error Probability



Topics to be Covered



Topic

Network Delays

Topic

Flow Control Protocol

ABOUT ME



Hello, I'm **Abhishek**

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[NAT]

IIT-R [GATE-2025, Set-1] [2 Mark]



#Q. Suppose a 5-bit message is transmitted from a source to a destination through a noisy channel. The probability that a bit of the message gets flipped during transmission is 0.01. Flipping of each bit is independent of one another. The probability that the message is delivered error-free to the destination is _____. (rounded off to three decimal places)

Message Size (L) = 5 bits
Bit Error Prob. (P_b) = 0.01

$$\begin{aligned} \text{Ans} &= (1 - P_b)^L \\ &= (1 - 0.01)^5 \\ &= 0.9509 \end{aligned}$$

IIT-R Ans =
0.949 to 0.952

Ans = 0.950 to 0.951

#Q. In a communication network, a packet of length L bits takes link $L1$ with a probability of p_1 or link $L2$ with a probability of p_2 . Link $L1$ and $L2$ have bit error probability of b_1 and b_2 respectively. The probability that the packet will be received without error via either $L1$ or $L2$ is

[GATE 2005]

☒ **A** $(1 - b_1)^L p_1 + (1 - b_2)^L p_2$

☐ **B** $[1 - (b_1 + b_2)^L] p_1 p_2$

☐ **C** $(1 - b_1)^L (1 - b_2)^L p_1 p_2$

☐ **D** $1 - (b_1^L p_1 + b_2^L p_2)$

$$P_1 * (1 - b_1)^L + P_2 * (1 - b_2)^L$$

Ans: A



Topic : Network Delays



Four types of network delays :

1. Transmission delay
2. Propagation delay
3. Queuing delay
4. Processing delay



Topic : Transmission Delay



- Transmission Time / Delay [t_x] (in seconds)
- Time required to transmit a packet over a link



$$\text{Transmission Delay} = \frac{[\text{Packet Size}]}{[\text{Data Transfer Rate}]}$$



Topic : Packet Size



- Packet Length or Frame Size
- Number of bits or bytes in one packet
- Size of Data (Digital) (Base-2)

$$1 \text{ KB} = 2^{10} \text{ bytes}$$

$$1 \text{ MB} = 2^{20} \text{ bytes}$$

$$1 \text{ GB} = 2^{30} \text{ bytes}$$

$$1 \text{ TB} = 2^{40} \text{ bytes}$$

B → Byte

b → Bits



Topic : Data Transfer Rate



→ Data Transfer Rate or Bandwidth

→ Number of bits or bytes transmitted per seconds = Bit Rate

→ Number or signals generated into channel per seconds ⇒ Baud Rate

→ Count or Frequency

→ Data Transfer Rate (Analog) (Base-10)

1 Kbps = 10^3 bits per second

1 Mbps = 10^6 bits per second

1 Gbps = 10^9 bits per second

1 Tbps = 10^{12} bits per second



Topic : Network Delays



→ 1 second = 10^3 milliseconds (ms)
= 10^6 microseconds (μ s)
= 10^9 nanoseconds (ns)
= 10^{12} picoseconds (ps)

	Analog	Digital
1 K	10^3	2^{10}
1 M	10^6	2^{20}
1 G	10^9	2^{30}

Example 1 :-

Consider frame size is 1000 bytes and bandwidth of a link is 1Mbps, then calculate transmission delay in milliseconds?

$$10^6 \text{ bits} \rightarrow 1 \text{ sec}$$

$$10^6 \text{ bits} \rightarrow 10^3 \text{ ms}$$

$$10^3 \text{ bits} \rightarrow 1 \text{ ms}$$

$$8 \times 10^3 \text{ bits} \rightarrow 8 \text{ ms}$$

$$1000 \text{ bytes} \rightarrow 8 \text{ ms}$$

Example 1 :-

Consider frame size is 1000 bytes and bandwidth of a link is 1Mbps, then calculate transmission delay in milliseconds?

Solution :

$$\text{Frame Size} = 1000 \text{ bytes} = 8 * 10^3 \text{ bits}$$

$$\text{Bandwidth} = 1 \text{ Mbps} = 10^6 \text{ bits / sec}$$

$$\underline{t_x} = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{8 * 10^3 \text{ bits}}{10^6 \text{ bits / sec}} = 8 * 10^{-3} \text{ sec}$$

$$= 8 \text{ ms}$$

$$\boxed{\text{Ans} = 8}$$

Example 2 :-

Consider packet size is 4 KB and data transfer rate of a link is 256 Kbps, then
Calculate transmission delay in milliseconds?

Example 2 :-

Consider packet size is 4 KB and data transfer rate of a link is 256 Kbps, then Calculate transmission delay in milliseconds ?

Solution :

$$\text{Frame Size} = 4 \text{ KB} = 2^{15} \text{ bits} = 2^2 * 2^{10} * 2^3 \text{ bits}$$

$$\text{Bandwidth} = 256 \text{ Kbps} = 2^8 * 10^3 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{2^{15} \text{ bits}}{2^8 * 10^3 \text{ bits / sec}} = 2^7 * 10^{-3} \text{ sec} = 128 \text{ ms}$$

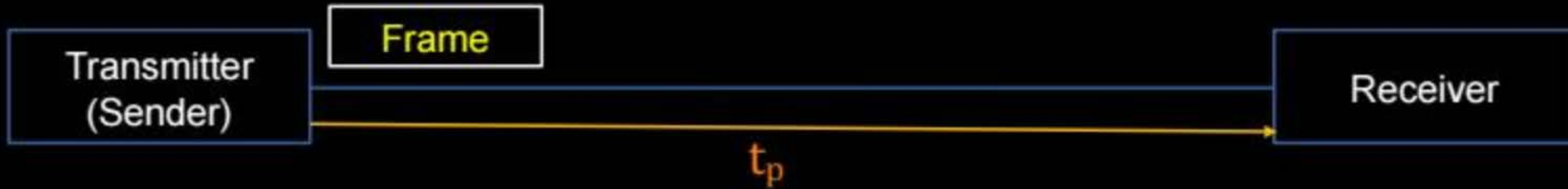
$$\boxed{\text{Ans} = 128}$$



Topic : Propagation Delay



- Propagation Time / Delay [t_p] (in seconds)
- Time required to travel a signal (bit) from one end to other end of a link
- One-way



$$\text{Propagation Delay} = \frac{[\text{Distance}]}{[\text{Signal Speed}]}$$

Example 3 :-

Consider distance between two host is 200 meter and signal speed of a link is 10^5 meter per second then calculate propagation delay in milliseconds?

$$10^5 \text{ m} \longrightarrow 1 \text{ sec}$$

$$10^5 \text{ m} \longrightarrow 10^3 \text{ ms}$$

$$200 \text{ m} \longrightarrow \frac{200}{10^5} \times 10^3 \text{ ms}$$

$$200 \text{ m} \longrightarrow 2 \text{ ms}$$

Example 3 :-

Consider distance between two host is 200 meter and signal speed of a link is 10^5 meter per second then calculate propagation delay in milliseconds ?

Solution :

$$\text{Distance} = 200 \text{ meter}$$

$$\text{Signal Speed} = 10^5 \text{ meter / sec}$$

$$t_p = \frac{\text{Distance}}{\text{Signal Speed}} = \frac{200 \cancel{\text{meter}}}{10^5 \cancel{\text{meter}} / \text{sec}} = 2 \times 10^{-3} \text{ sec} = 2 \text{ ms}$$

$$\boxed{\text{Ans} = 2}$$

Example 4 :-

Consider distance between two host is 2 Km and signal speed of a link is 5 microsecond per Km then calculate propagation delay in microseconds ?

$$t_p = \frac{\text{Distance}}{\text{Signal speed}} = \frac{2 \text{ Km}}{5 \mu\text{s/Km}}$$

1 Km $\longrightarrow$ 5 μs

2 Km $\longrightarrow$ 2 * 5 μs

10 μs

Example 4 :-

Consider distance between two host is 2 Km and signal speed of a link is 5 microsecond per Km then calculate propagation delay in microseconds ?

Solution :

$$\begin{aligned} \text{Distance} &= 2 \text{ Km} \\ \text{Signal Speed} &= 5 \mu\text{s} / \text{Km} \end{aligned}$$

$$t_p = \text{Distance} * \text{Signal Speed} = 2 \text{ Km} * 5 \mu\text{s} / \text{Km} = 10 \mu\text{s}$$

$$\text{Ans} = 10$$



Topic : Propagation Delay



→ if signal speed given in "meter per second"

$$\text{Propagation Delay} = \frac{\text{Distance}}{\text{Signal Speed}}$$

→ if signal speed given in "second per meter"

$$\text{Propagation Delay} = \text{Distance} \times \text{Signal Speed}$$



Topic : Round Trip Propagation Delay



→ Two-way

→ 2 * Propagation Time $[2 * t_p]$

#Q. Consider two hosts X and Y, connected by a single direct link of rate 10^6 bits/sec. The distance between the two hosts is 10,000 km and the propagation speed along the link is 2×10^8 m/s. Hosts X send a file of 50,000 bytes as one large message to hosts Y continuously. Let the transmission and propagation delays be p milliseconds and q milliseconds, respectively. Then the values of p and q are:

- (A) $p = 50$ and $q = 100$
- (B) $p = 50$ and $q = 400$
- (C) $p = 100$ and $q = 50$
- (D) $p = 400$ and $q = 50$

[GATE-2017, Set-2, MCQ, 2-Mark]

IIT-R, H.W.



Topic : End-to-End Delay



- One-way delay
- Time required for a packet to be transmitted from Transmitter to Receiver

End-to-end delay = Transmission delay + Propagation delay

$$= (t_x + t_p)$$



Sender
Host A

Receiver
Host B



1) Transmit Window (Buffer)

2) Receiving Window

Transmitter (Sender)
Transmitting (Sending)
Buffer (Window)

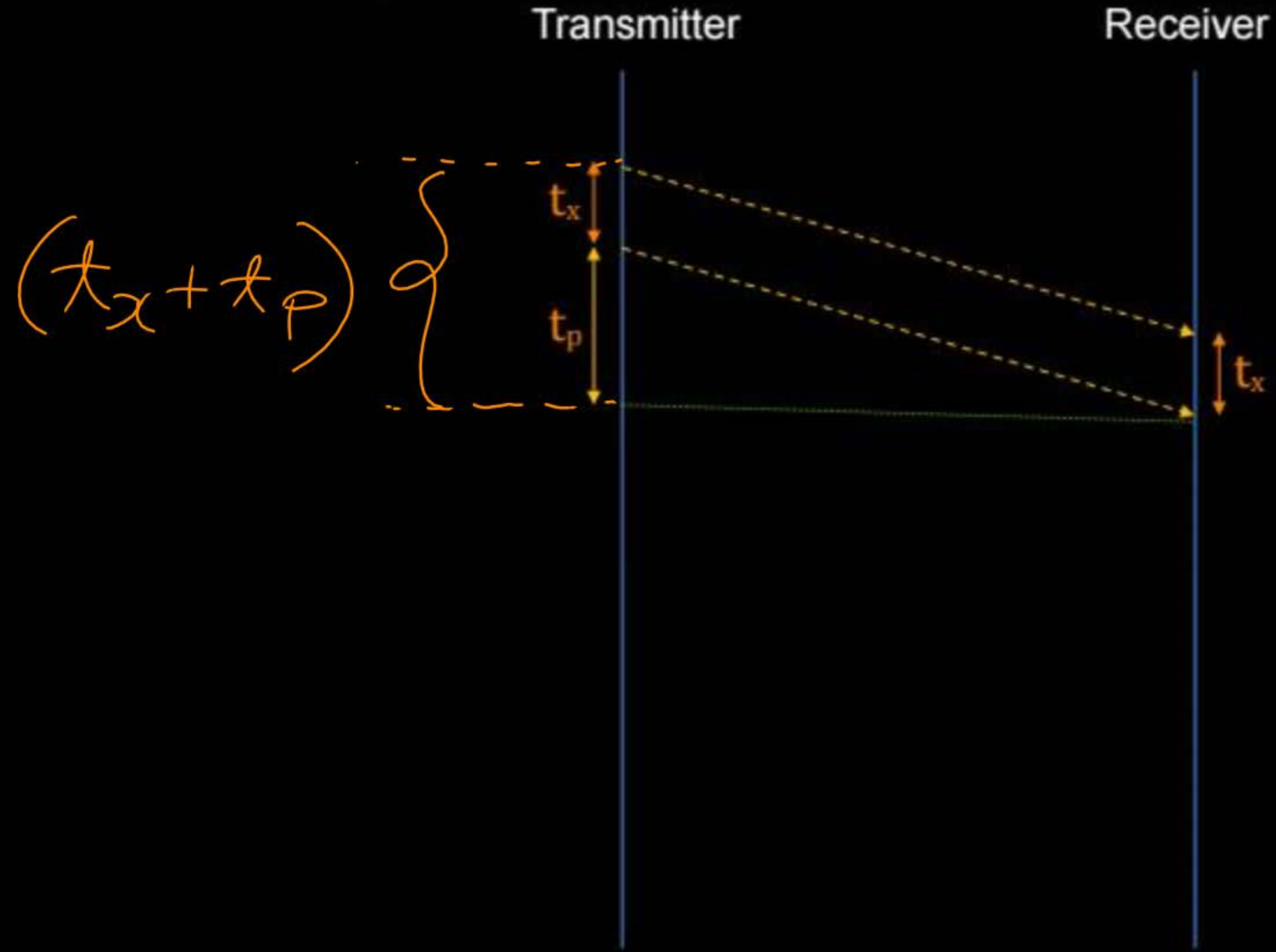
1) Transmit window

2) Receiving Window

Receiver
Receiving
Window



Topic : End-to-End Delay



#Q. Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at a speed of 3×10^8 m/s. The time taken (in milliseconds, rounded off to two decimal places) for the receiver to completely receive a packet of 1000 bytes transmitted by the sender is _____.

[GATE-2022, NAT, 2-Marks]
IIT-KGP, H.W.



Topic : Network Delays



CASE I :

Transmission delay < Propagation delay



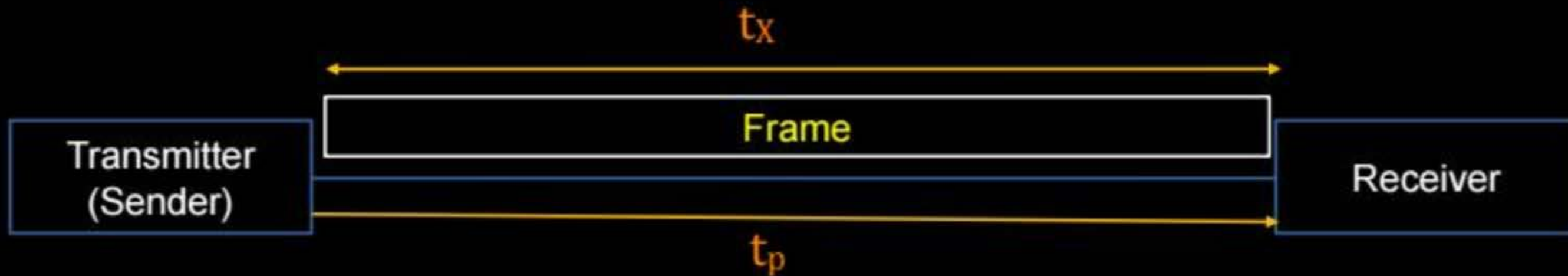


Topic : Network Delays



CASE II :

Transmission delay = Propagation delay



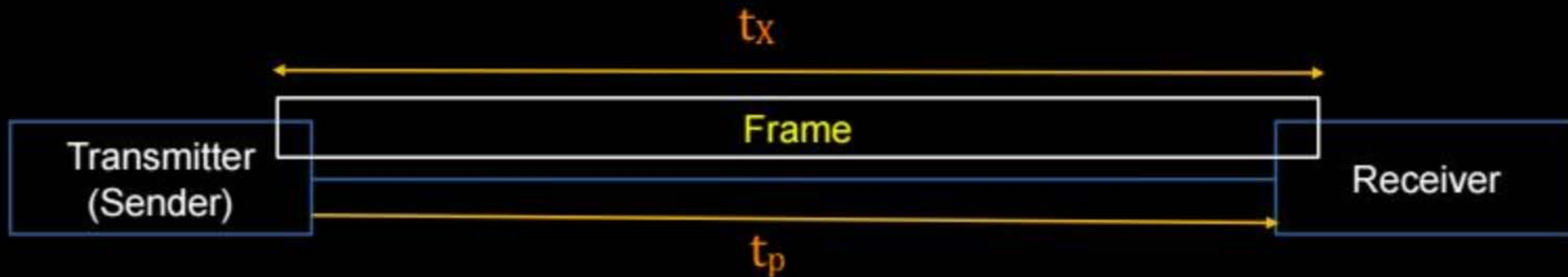


Topic : Network Delays



CASE III :

Transmission delay $>$ Propagation delay





Topic : Queuing Delay



- Waiting time of a packet at input buffer, before processing
- Cannot be determined
- if not given, consider negligible



Topic : Processing Delay



- Time required to process a packet after receiving
- Based on CPU processing speed and packet size
- if not given, consider negligible



2 mins Summary



Topic

Network Delays ✓

Topic

~~Stop-and-Wait ARQ~~



THANK - YOU

